Real Numbers

Real numbers are the numbers that can be represented on the number line. They include the whole numbers, their opposites, and all the other numbers in between them.

The real numbers are a union of the rational and irrational numbers.

Rational Numbers - Rational numbers are numbers that can be written as a quotient of two integers.

The natural numbers, whole numbers, and integers are subsets of the rational numbers. Note: A member of any of these sets can be expressed as a quotient of two integers.

Natural Numbers: {1, 2, 3, 4, 5...}

Whole Numbers: {0, 1, 2, 3, 4, 5...}

Integers: {...–5, –4, –3, –2, –1, 0, 1, 2, 3, 4, 5...}

Also included in the rational numbers are **fractions** (where the numerator and denominators are integers) and **repeating and terminating decimals**. Fractions may not have 0 as the denominator.

Irrational Numbers – Irrational numbers are numbers that cannot be expressed as the ratio of two integers. Examples of irrational numbers are (a) square roots of non-perfect squares, (b) pi (π), and (c) decimals that do not develop into a repeating pattern.

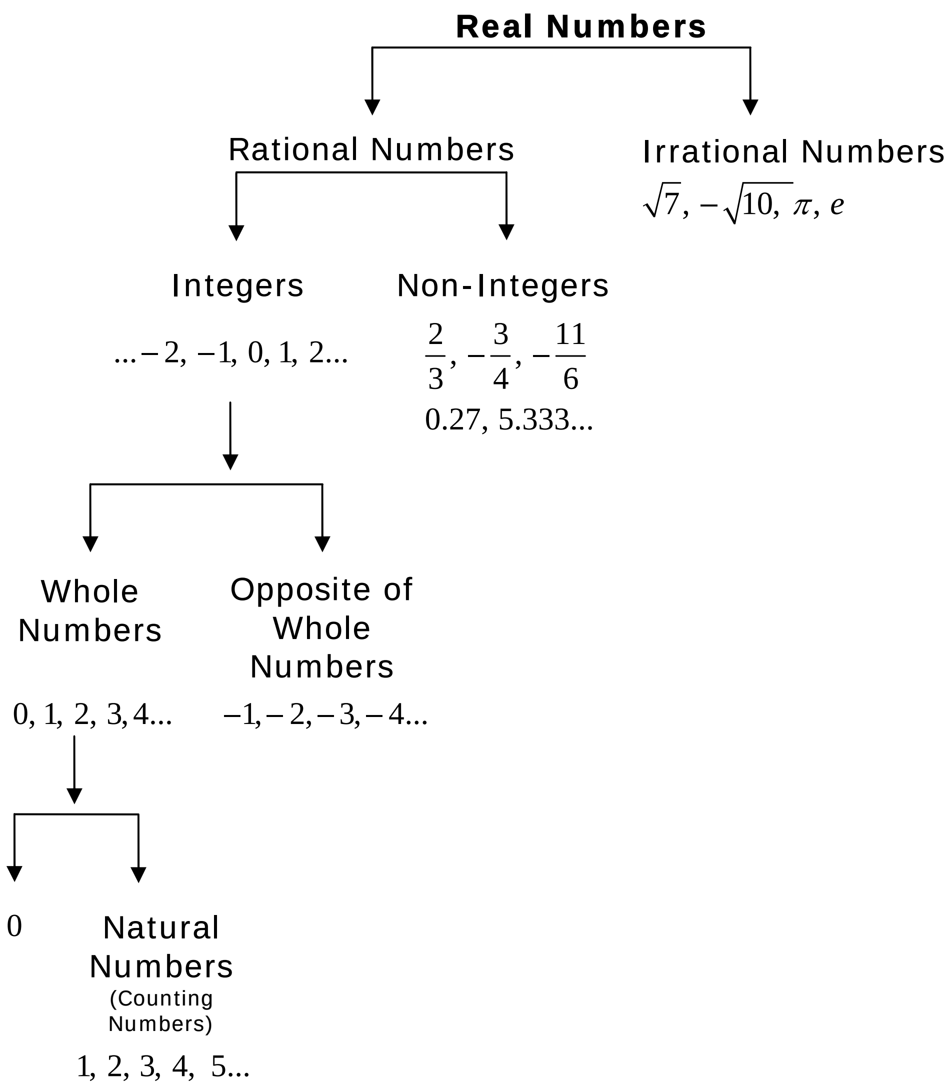
(a) $\sqrt{94}$
(b) $\pi \approx$ 3.1415926535897932384626433832795...
(c) 9.6953597148326580281488811508453

real numbers = rational numbers + irrational numbers

Study the two sketches below to better understand the subsets of the real numbers.

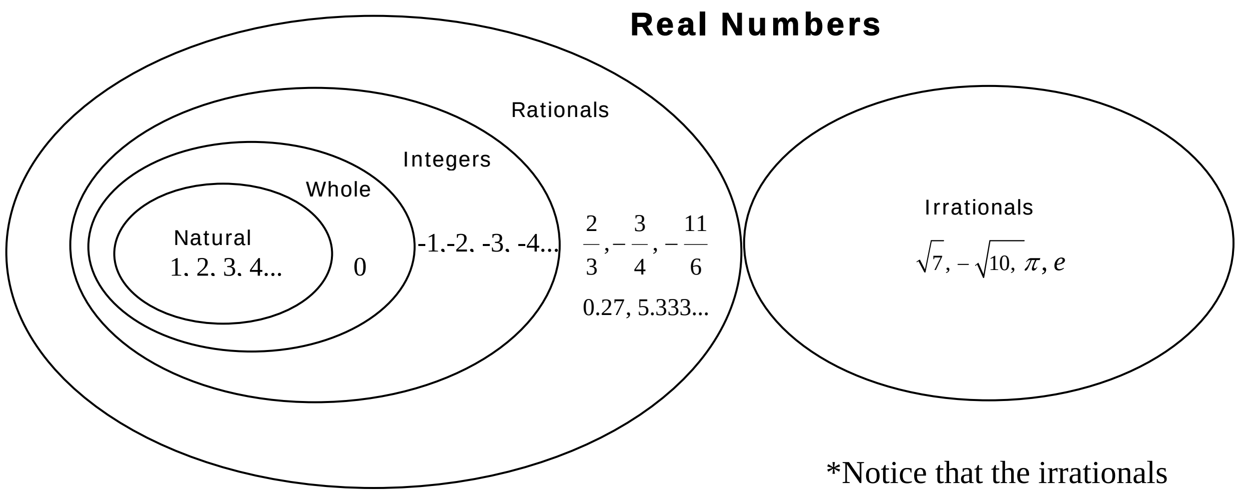
The first sketch shows how each subset can be divided into further subsets. For example, 0 is a whole number, but is also an integer and a rational number because it is a member of all those subsets. The only subset of the rational numbers that 0 is not a part of is the natural numbers.

The second sketch shows the connection between the subsets using a Venn diagram. Again, looking at zero, it is in all of the ovals of the rational numbers with the exception of the natural numbers.



* The numbers listed in the subsets are just a few examples of the infinite possibilities.

The next figure shows the connection between the subsets of real numbers using a Venn diagram.



*Notice that the irrationals are in a league of their own.

The **rational numbers** are the types of numbers shown within the “rational circle”. The “rational circle” includes all of the numbers within the circles inside the it. Some examples of rational numbers are $\left(\frac{-11}{6}\right)$, $\left(-4=\frac{-4}{1}\right)$, $\left(0=\frac{0}{1}\right)$, and $\left(4=\frac{4}{1}\right)$.

The **integers** are the types of numbers shown within the “integer circle”. The “integer circle” includes all of the numbers within the circles inside it. Some examples of integers are $-4, 0, 4$.

The **whole numbers** are the types of numbers shown within the “whole number circle”. The “whole number circle” includes all of the numbers within the circle inside it. Some examples of whole numbers are 0 and 4.

The **natural numbers** are the types of numbers shown within the “natural number circle”. Some examples of whole numbers are 3 and 4.

The **irrational numbers** are in a “league of their own”. Some examples of rational numbers are $\sqrt{26}$, $-\sqrt{43}$, π, and e.

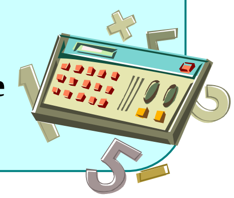
The **real numbers** are the types of numbers shown within all of the circles. The real numbers are the union of the rational and the irrational numbers. Some examples of real numbers are $\left(\frac{-11}{6}\right)$, $\left(-4=\frac{-4}{1}\right)$, $\left(0=\frac{0}{1}\right)$, and $\left(4=\frac{4}{1}\right)$; and also $\sqrt{26}$, $-\sqrt{43}$, π, and e.

Rational and Irrational Numbers

Calculators can help us understand the difference between **rational** and **irrational** numbers.

Rational numbers are numbers that can be written as fractions, and when expressed as decimals are either a repeating or terminal decimal. **Irrational numbers** are numbers that have decimals that go on forever, but never develop a repeating pattern.

To express a fraction as a decimal, divide the numerator (top number) by the denominator (bottom number).



Definition	Type of Decimal	Calculator Buttons
Rational numbers are numbers that **can** be expressed as a fraction where both the numerator and the denominator are integers (with the exception of 0 as a denominator*).	**Repeating** $\frac{5}{11} = \mathbf{0.454545...}$ **Terminating** $\frac{7}{8} = \mathbf{0.875}$	**5** **÷** **1** **1** **=** **7** **÷** **8** **=**
Irrational numbers are numbers that **cannot** be expressed as a fraction where both the numerator and the denominator are integers.	**Do not repeat or terminate** **First 12 digits of** $\pi(pi)$ **= 3.14159265359** **First 7 digits of** $\sqrt{2} = \mathbf{1.414213}$	**Not all calculators use the same symbols, so the "pi" and "square root" buttons may look a little different.** π $\sqrt{\ }$ **2**

** Note: Since division by zero is undefined, the denominator of a fraction cannot equal 0.*

Using a calculator, we can explore irrational and rational numbers.

$\frac{2}{3}$ = 2 ÷ 3 = 0.6666666 – repeating decimal – **rational**

$\sqrt{50}$ = square root ($\sqrt{\ }$) of 50 = 7.0710678 – non-repeating decimal – **irrational**

$\frac{4}{5}$ = 4 ÷ 5 = 0.8 – terminating decimal – **rational**

Properties of Real Numbers

In the previous section you learned how to translate verbal phrases into algebraic sentences. You were able to do this because of mathematical properties. In this section you will study these properties that you will use in future units to solve equations.

Commutative Properties of Addition and Multiplication	
The **order** in which numbers are added does not make a difference in the sum. $6 + 4 = 4 + 6$ For any numbers x and y, $x + y = y + x$	The **order** in which numbers are multiplied does not make a difference in the product. $6 \times 4 = 4 \times 6$ For any numbers x and y, $xy = yx$

Associative Properties of Addition and Multiplication	
The way in which numbers are **grouped** does not change the sum. $(2 + 3) + 4 = 2 + (3 + 4)$ For any numbers x, y, and z, $(x + y) + z = x + (y + z)$	The way in which numbers are **grouped** does not change the product. $(2 \times 3) \times 4 = 2 \times (3 \times 4)$ For any numbers x, y, and z, $(xy)z = x(yz)$

Identity Properties of Addition and Multiplication	
The sum of a number and zero is that number. $3 + 0 = 3$ For any number n, $n + 0 = n$	The product of a number and one is the number. $5 \cdot 1 = 5$ For any number n, $n \cdot 1 = n$

Multiplicative Property of Zero

The product of a number and zero is zero.

$7 \cdot 0 = 0$

For any number n, $n \cdot 0 = 0$

These properties will be helpful when you start solving equations later on in the course. Right now you will be asked in the assignment to identify properties illustrated by algebraic expressions, so let's practice deciding which property is shown.

Examples:

1) $4 + (9 + 5) = (4 + 9) + 5$ Associative Property of Addition

2) $y + 14 = 14 + y$ Commutative Property of Addition

3) $(9 + n) + 6 = 6 + (9 + n)$ Commutative Property of Addition

4) $5m \cdot 0 = 0$ Multiplicative Property of Zero

5) $0 + 17 = 17$ Identity Property of Addition

6) $8xyz \cdot 1 = 8xyz$ Identity Property of Multiplication